Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

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November 9, 2026

1. **Consider the matrices**

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 0 & 1 & 1 \\ -1 & 1 & 2 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 & 1 & 1 \\ -2 & -4 & -6 & -2 \\ -2 & -1 & -1 & 0 \end{bmatrix}.$$

a) **Determine elementary matrices H, J and K such that**

$$HA = B, \qquad JA = C, \qquad KA = D.$$

$$H = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad R_1 \leftrightarrow R_2, \qquad HA = B.$$

$$J = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad R_2 \leftarrow R_2 + R_3, \qquad JA = C.$$

$$K = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad R_2 \leftarrow -2R_2, \qquad KA = D.$$

b) **Determine, if they exist, elementary matrices $E_p, \ldots, E_1$ such that**

i)

$$E_p \cdots E_1 A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 2 & 2 & 0 \end{bmatrix}.$$

$$A \xrightarrow[t_1]{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & 2 & 0 \\ -2 & -1 & -1 & 0 \end{bmatrix} \xrightarrow[t_2]{R_3 \leftarrow R_3 + 2R_1} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & 2 & 0 \\ 0 & -1 & 1 & 2 \end{bmatrix}$$

$$\xrightarrow[t_3]{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 2 & 2 & 0 \end{bmatrix}.$$

The corresponding elementary matrices are

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Hence,

$$\boxed{E_3 E_2 E_1 A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 2 & 2 & 0 \end{bmatrix}}.$$

ii)

$$E_p \cdots E_1 A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 3 & -3 & -6 \\ 0 & 0 & 7 & 7 \end{bmatrix}.$$

We reuse the first three elementary transformations from i):

$$E_3 E_2 E_1 A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 2 & 2 & 0 \end{bmatrix}$$

$$\xrightarrow[t_4]{R_2 \leftarrow -3R_2} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 3 & -3 & -6 \\ 0 & 2 & 2 & 0 \end{bmatrix} \xrightarrow[t_5]{R_3 \leftarrow R_3 - \frac{2}{3}R_2} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 3 & -3 & -6 \\ 0 & 0 & 4 & 4 \end{bmatrix}$$

$$\xrightarrow[t_6]{R_3 \leftarrow \frac{7}{4}R_3} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 3 & -3 & -6 \\ 0 & 0 & 7 & 7 \end{bmatrix}.$$

The additional elementary matrices are

$$E_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix}, \quad E_6 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{7}{4} \end{bmatrix}.$$

Therefore,

$$\boxed{E_6 E_5 E_4 E_3 E_2 E_1 A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 3 & -3 & -6 \\ 0 & 0 & 7 & 7 \end{bmatrix}}.$$

c) **Determine a matrix Qsuch that $QA = F$, where**

$$F = \begin{bmatrix} -2 & 4 & 2 & -2 \\ 0 & 2 & 2 & 0 \\ 0 & 3 & 9 & 6 \end{bmatrix}.$$

We apply elementary row operations to transform A into F.

$$A \xrightarrow[t_1]{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & 2 & 0 \\ -2 & -1 & -1 & 0 \end{bmatrix} \xrightarrow[t_2]{R_3 \leftarrow R_3 + R_2} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & 2 & 0 \\ -2 & 1 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow[t_3]{R_3 \leftarrow R_3 + 2R_1} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & 2 & 0 \\ 0 & 1 & 3 & 2 \end{bmatrix} \xrightarrow[t_4]{R_3 \leftarrow 3R_3} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & 2 & 0 \\ 0 & 3 & 9 & 6 \end{bmatrix}$$

$$\xrightarrow[t_5]{R_1 \leftarrow R_1 - R_2} \begin{bmatrix} 1 & -2 & -1 & 1 \\ 0 & 2 & 2 & 0 \\ 0 & 3 & 9 & 6 \end{bmatrix} \xrightarrow[t_6]{R_1 \leftarrow -2R_1} \begin{bmatrix} -2 & 4 & 2 & -2 \\ 0 & 2 & 2 & 0 \\ 0 & 3 & 9 & 6 \end{bmatrix} = F.$$

The corresponding elementary matrices are

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix},$$

$$E_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \quad E_5 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_6 = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

By associativity of matrix multiplication,

$$E_6 E_5 E_4 E_3 E_2 E_1 A = F.$$

Therefore, taking

$$\boxed{Q = E_6 E_5 E_4 E_3 E_2 E_1 = \begin{bmatrix} -4 & 2 & 0 \\ -1 & 1 & 0 \\ 3 & 3 & 3 \end{bmatrix}}$$

we obtain $\boxed{QA = F}$.

Key idea of this exercise:

$$\underbrace{A \xrightarrow{E_1} A_1 \xrightarrow{E_2} A_2 \xrightarrow{E_3} \cdots \xrightarrow{E_p} B}_{\text{Elementary row operations}}$$

$$\Updownarrow$$

$$\boxed{B = E_p \cdots E_2 E_1 A}$$